



*“There are grounds for cautious optimism
that we may now be near the end of the
search for the ultimate laws of nature.”*

Stephen Hawking

THE UNIVERSE IS ALL OF EXISTENCE—
all of space and time and all the matter and energy
within it. The universe is unknowably vast, and ever
since it formed, it has been expanding, carrying
distant regions apart at speeds up to—and in some
cases exceeding—the speed of light. The universe
encompasses everything from tiny subatomic
particles to vast galaxy superclusters, yet it seems
that all are governed by the same basic laws. All
visible matter (which is only a small percentage of
the total matter) is built from the same subatomic
blocks, and the same fundamental forces govern
all interactions between these elements. Knowledge
of these cosmic operating principles—from general
relativity to quantum physics—informs cosmology,
the study of the universe as an entity. Cosmologists
hope to answer questions such as “How big is
the universe?”, “How old is it?”, and “How does
it work, on the grandest scale?”

BOW SHOCK AROUND A STAR

This mysterious image from the Orion Nebula shows how matter and radiation interact on a stellar scale. A star surrounded by gas and dust has met a fierce wind of particles blowing from a bright young star (out of frame). Around the star, a crescent-shaped gaseous bow shock has built up, like water flowing past the prow of a boat.

WHAT IS THE UNIVERSE?